Quiz - 1
20/07/2025

Class - 7

Impedence

The impedance z of a circuit is the ratio of the phasor current I, measured in ohm(Ω). The impedence represents the opposition that the circuit exhibits to the flow of sinusoidal current

$$Z = \frac{V}{I} = R + jX = |Z| \angle \theta$$

The admittance Y is the reciprocal of ~~inde~~ impedance measured in Siemens(S) or ohms.

$$Y = \frac{1}{Z} = \frac{I}{V} = G + jB \rightarrow \text{Susceptance}$$

$\hookrightarrow$ conductance. | $G = \frac{1}{R}$ wl | $X = 0$ ohi

element	Impedance	Admittance
R	R	1/R
L	jwL	1/jwL
C	$\frac{1}{jwC}$	jwC

In DC → when, $\omega = 0$ → $j\omega L = 0$ → means short circuit

$\frac{1}{j\omega L} = \alpha$ → means open circuit

In AC → frequency exists.

If frequency is very high → $j\omega L$ = high → means open circuit.

$\frac{1}{j\omega L}$ = low → means short circuit.

In usual frequency, it behaves like DC circuits type

Phasor diagram

phasor diagram shows at a glance the magnitudes and phase relations, among the various quantities (current or voltage) within the network, in the complex number plane phase angles are measured counter clockwise from the positive real axis, and magnitudes are measured from the origin of the axes.

#. Two different magnitude scales are necessary. one for currents and one for voltages.

#. Both must have common angle scaling.

Resistive elements

$v_R(t) = V_m \sin\omega t = V_m\angle 0^\circ$

$I_R(t) = I_m \sin\omega t = I_m\angle 0^\circ$

$\Rightarrow Z_R = \frac{V_R}{I_R} = \frac{V_m\angle 0^\circ}{I_m\angle 0^\circ}$

$\Rightarrow \boxed{Z_R = R\angle 0^\circ}$

$Z_R = R$

↳ reactangular form.

Inductive Reactance

$v_L(t) = V_m \sin(\omega t) = V_m\angle 0^\circ$

$i_L(t) = I_m \sin(\omega t - 90^\circ)$

$= I_m\angle -90^\circ$

$Z_L = \frac{V_L}{i_L}$

$\Rightarrow Z_L = \frac{V_m\angle 0^\circ}{I_m\angle -90^\circ}$

$\Rightarrow \boxed{Z_L = X_L\angle 90^\circ}$

↳ $X_L = \omega L$

$\therefore Z_L = j\omega L$

↳ rectangular form.

capacitive Reactance.

$v_c(t) = V_m \sin(\omega t) = V_m\angle 0^\circ$

$i_c(t) = I_m \sin(\omega t + 90^\circ) = I_m\angle 90^\circ$

$Z_c = \frac{V_c}{I_c} = \frac{V_m\angle 0^\circ}{I_m\angle 90^\circ}$

$\boxed{Z_c = X_c\angle -90^\circ}$

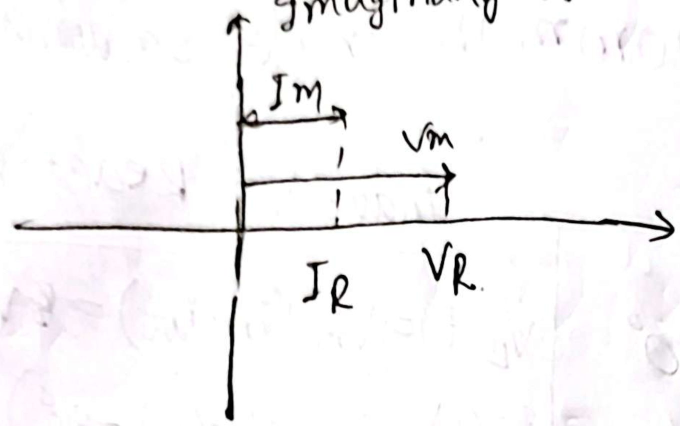
$X_c = \frac{1}{\omega C}$

$\therefore Z_c = -jX_c$ → reptangular form.

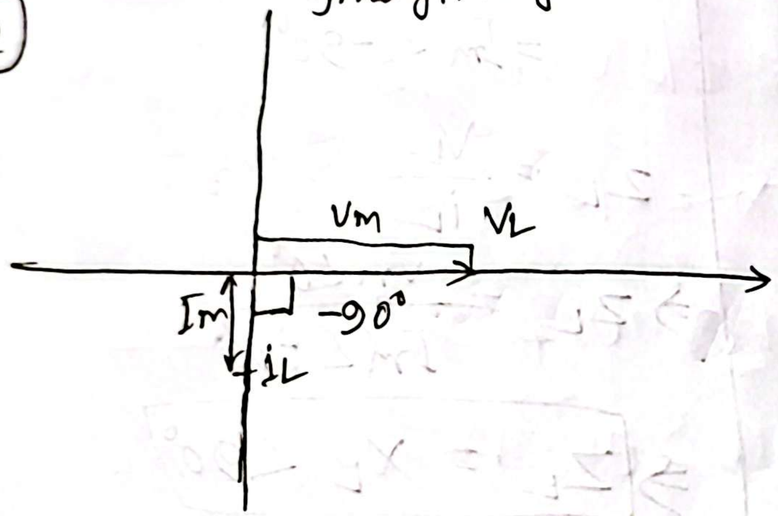
~~Imaginary element~~ \ element circuit.

Phasor Diagram

①



②

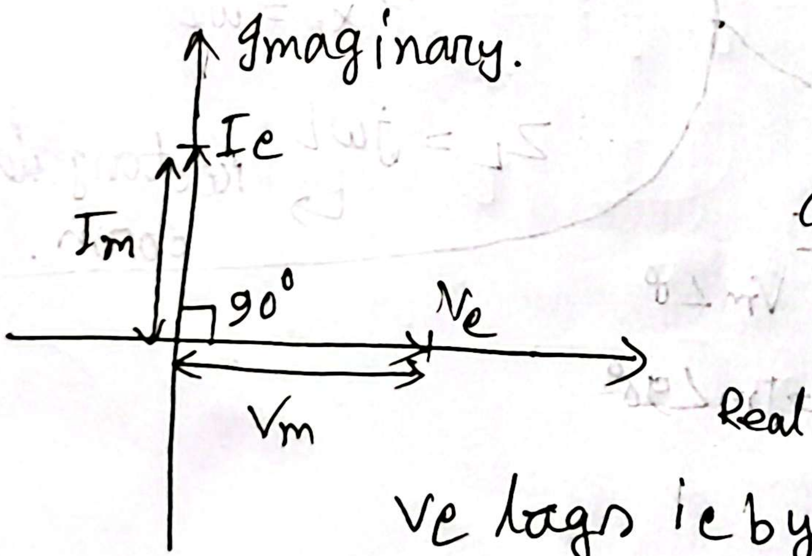


If the angle gone into counter clockwise (we have to go through short angle)

V_L leads i_L by 90^0

i_L lags V_L by 90^0.

3)



counter clockwise → lag

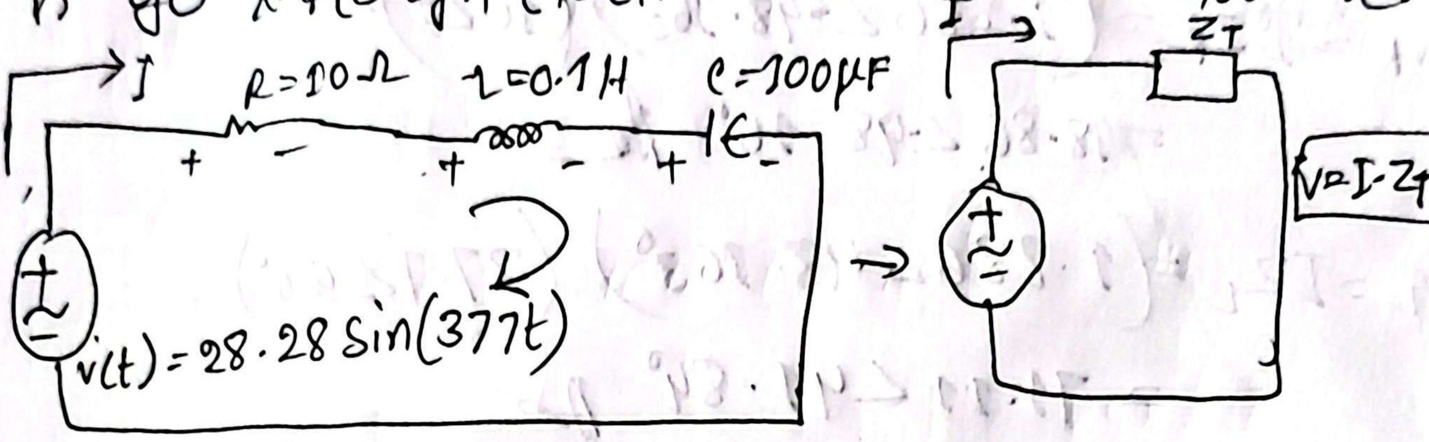
clockwise → lead.

V_c lags i_c by 90^0

(V_c / I_c)

we have to assume one variable as a main term, Then, we have to see if it is go through clockwise or Anti-clockwise.

a]



a] calculate I, V_R, V_L and V_C in phasor form.

Solve

$$Z_R = R\angle 0^\circ = 10\angle 0^\circ$$

$$Z_L = X_L\angle 90^\circ = \omega L\angle 90^\circ = 377\times 0.1\times \angle 90^\circ$$
$$= 37.7\angle 90^\circ\ \Omega$$

$$Z_C = X_C\angle -90^\circ = \frac{1}{\omega C} = \frac{1}{377\times 100\times 10^{-6}}\angle -90^\circ$$
$$= 26.53\angle -90^\circ\ \Omega$$

$V = 28.28\angle 0^\circ$ volt.

Applying KVL, $-V + V_R + V_L + V_C = 0$

$$\Rightarrow V = V_R + V_L + V_C$$

$$\Rightarrow V = IZ_R + IZ_L + IZ_C$$

$$\Rightarrow V = I(Z_R + Z_L + Z_C)$$

$$\Rightarrow I = \frac{V}{Z_R + Z_L + Z_C}$$

$$= \frac{28.28\angle 0^\circ}{10\angle 0^\circ + 37.7\angle 90^\circ + 26.53\angle -90^\circ}$$

$$\Rightarrow I = 1.886\angle -48.163^\circ A$$

$$V_R = I Z_R = (1.886\angle -48.163^\circ)(10\angle 0^\circ)$$

$$= 18.86\angle -48.16^\circ$$ Ans

$$V_L = I Z_L = (1.886\angle 48.163^\circ)(37.7\angle 90^\circ)$$

$$= 71.11\angle 41.84^\circ$$ Ans

$$V_C = I Z_C = (1.886\angle -48.16^\circ)(26.253\angle -90^\circ)$$

$$= 50.03\angle -138.16^\circ$$ Ans

$$Z_T = Z_R + Z_L + Z_C$$

$$= 15\angle 48.16^\circ\,\Omega$$ Ans